Experiment No:6 Parseval's Relation and Harmonic Spectrum Analysis

Objective:

(i)To verify Parseval's relation for continuous-time periodic signals and analyze the relationship between time-domain and frequency-domain power.

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Procedure:

• Generate the periodic signal over one period.

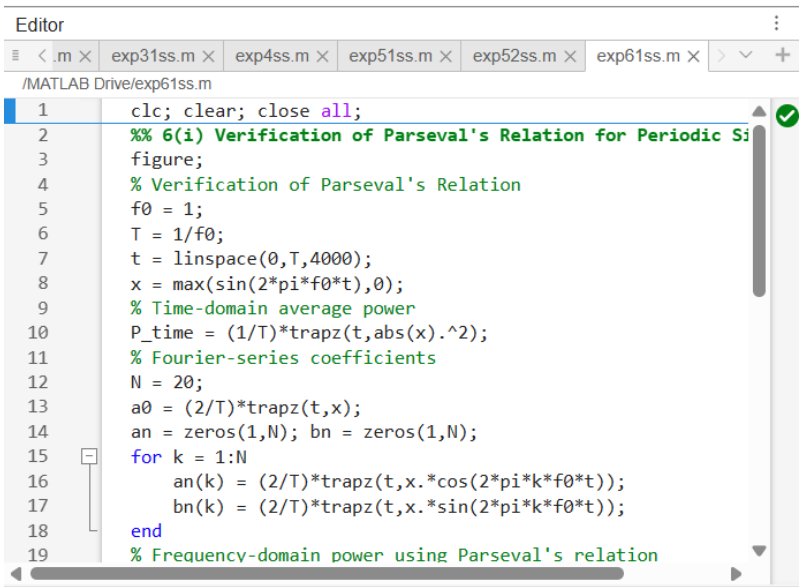
• Compute time-domain average power.

• Calculate Fourier coefficients up to N = 20.

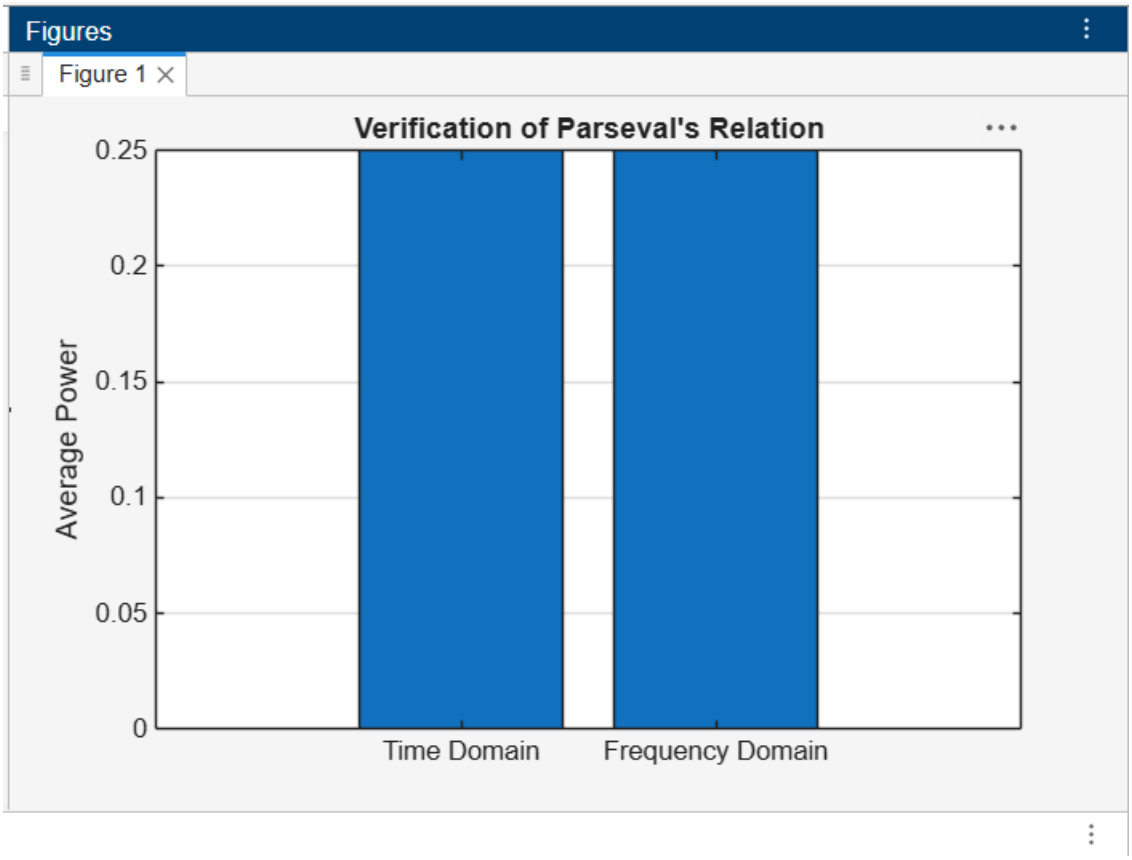
• Evaluate frequency-domain power using Parseval's relation.

• Compare the two values and calculate absolute error.

Matlab Coding:



Output:



(iv) To compare the total power obtained in the time domain with the harmonic power distribution in the frequency domain.

Experimental Procedure

• Generate the periodic signal and compute Fourier coefficients.

• Calculate harmonic magnitude for n = 1 to 15.

• Convert harmonic number to frequency.

• Plot the harmonic spectrum using a stem plot.

• Identify the significant components.

MATLAB Coding:

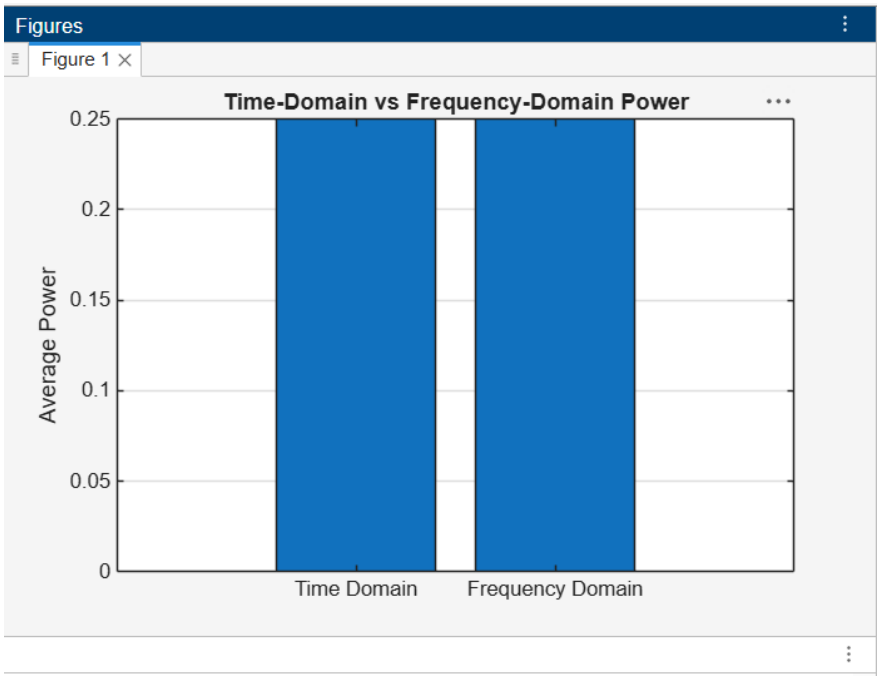
Editor

exp4ss.m | exp51ss.m | exp52ss.m | exp61ss.m | exp64ss.m

/MATLAB Drive/exp64ss.m

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clc; clear; close all;
%% 6(iv) Comparison of Time-Domain Power with Frequency-Dom
figure;
% Time-Domain and Frequency-Domain Power Comparison
f0 = 1; T = 1/f0;
t = linspace(0,T,4000);
x = max(sin(2*pi*f0*t),0);
P_time = (1/T)*trapz(t,x.^2);
N = 20;
a0 = (2/T)*trapz(t,x);
an = zeros(1,N); bn = zeros(1,N);
for k = 1:N
    an(k) = (2/T)*trapz(t,x.*cos(2*pi*k*f0*t));
    bn(k) = (2/T)*trapz(t,x.*sin(2*pi*k*f0*t));
end
P_dc = a0^2/4;
P_harmonics = 0.5*sum(an.^2+bn.^2);
P_frequency = P_dc + P_harmonics;
fprintf('Time-domain power = %.6f\n',P_time);
```

Output:



Conclusion:

The experiment successfully verified Parseval's relation for continuous-time periodic signals by demonstrating the equivalence of power calculated in the time and frequency domains.Thedistribution of power among the fundamental and harmonic components was analyzed through the harmonic spectrum, and the total time-domain power was found to be consistent with the sum of harmonic powers in the frequency domain.